



Donggeon Bae
Undergraduate Researcher



01 January 2000
Seoul, Korea
+01 3236-2599
www.d-bae.com/
<https://github.com/AttiBae>
atti2599@yonsei.ac.kr

About me

I'm a challenging and curious researcher. I have a strong interest in various fields of Machine Learning and their applications, particularly aspiring to extend into the bio-medical domain. I'm always looking forward to exchanging ideas with everyone for making the world a fancy place alongside me.

Education

- 2018 - 2024 BS. candidate in ECE (Expected) Yonsei University
GPA: 3.80/4.3(total), 3.93/4.3(major)
- 2015 - 2017 High school Poongsan Highschool
Specializing in mathematics and physics.

Research Interest

Computer Vision, Bio-medical Applications, Computational Imaging

Research Experiences

- 2020 - 2024 **Optical Imaging System Laboratory, Yonsei University**
Full-time Undergraduate Researcher, Supervisor: [Prof. Seung Ah Lee](#)
- Lensless Reconstruction With Deep Neural Networks
 - Deep Learning for Comprehensive Lensless Forward Modeling
 - Laser Speckle Contrast Imaging With Deep Neural Networks

Conference Paper

- 2020 - 2024 **"Lensless Imaging with an End-to-End Deep Neural Network"**
Donggeon Bae, Jaewoo Jung, Nakkyu Baek and Seung Ah Lee IEEE

Presentation

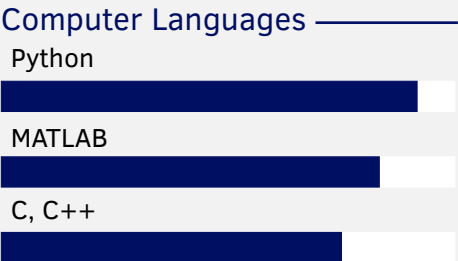
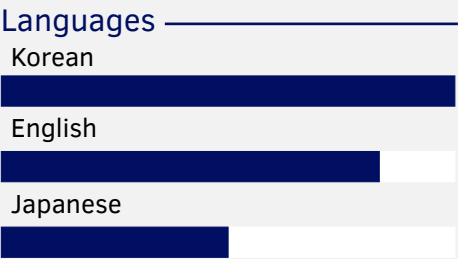
- 2023 **Poster, Optics and Photonics Congress**
"Learned Spatially Varying Forward Model for Lensless Imaging", **Donggeon Bae**, Jaewoo Jung, Nakkyu Baek, Taeyoung Kim, Kyungchul Lee and Seung Ah Lee Best Poster Award
- 2023 **Poster, Optics and Photonics Congress**
"Single-shot fabrication of light modulating phase mask for lensless camera using off-the-shelf components" Nakkyu Baek, Kyung Chul Lee, Junghyun Bae, Joonsik Park, Taeyoung Kim, **Donggeon Bae**, Wook Park, and Seung Ah Lee Best Poster Award
- 2023 **Poster, Optics and Photonics Congress**
"Robust Lensless Image Reconstruction via Forward Model Constraint Learning" Qianqian Cheng, Jaewoo Jung, **Donggeon Bae**, Lianfa Bai, Jing Han and Seung Ah Lee
- 2022 **Oral, IEEE Region 10 Conference**
"Image Reconstruction in Lensless Cameras with Unrolled Optimization Algorithms", Jaewoo Jung, Tae Young Kim, **Donggeon Bae**, Ernest K. Ryu and Seung Ah Lee
- 2022 **Poster, Gordon Research Conference Image Science**
"Deep Learning Approaches for Image Reconstruction in Lensless Cameras", Jaewoo Jung, **Donggeon Bae**, Kyungchul Lee, Nakkyu Baek, Ernest K. Ryu and Seung Ah Lee
- 2021 **Oral, IEEE/IEIE ICCE-ASIA**
"Image Reconstruction in Lensless Cameras with Unrolled Optimization Algorithms", Jaewoo Jung, Tae Young Kim, **Donggeon Bae**, Ernest K. Ryu and Seung Ah Lee
- 2021 **Poster, IEEE/IEIE ICCE-ASIA**
"Unrolled ADMM for Image Reconstruction in Lensless Cameras", Jaewoo Jung, Tae Young Kim, **Donggeon Bae**, Ernest K. Ryu and Seung Ah Lee
- 2020 **Oral, IEEE/IEIE ICCE-ASIA**
"Lensless Imaging with an End-to-End Deep Neural Network", **Donggeon Bae**, Jaewoo Jung, Nakkyu Baek and Seung Ah Lee

Donggeon Bae

Undergraduate Researcher

Machine Learning Expertise —

I have dedicated much time to implementing various forms of machine learning architectures, specially in area of CV. Additionally, I possess the ability to design and implement models that incorporate physics models of specific applications, as well as generating models such as Diffusion and GAN.



Scale: 0(Basic)-MAX(Commercial software production)

Awards

- 2023 Semester Honors (Top 10%)
- 2023 Best Poster Award at Optics and Photonics Congress

Extracurricular Activities

- 2021 - 2024 YAI, Yonsei Artificial Intelligence Yonsei University
 - Developed Dataset Generation Program for Detection Model
 - Implemented Cell Segmentation
 - Implemented Deep Learning Methods for Medical Diagnosis
 - Reviewed Generative Model Papers
- 2020 - 2022 Korean Millitary Service Sergeant discharge upon completion
- 2019 - 2020 FLOW, Yonsei Student Council Yonsei University
 - Worked as a member of General Affairs Department

Skills

- Software Pytorch, Tensorflow, Linux, VTK/ITK, Latex, CAD(Autodesk Fusion 360)
- Experiment Optical System Setup, Arduino, Raspberry Pi, 3D Printing
- ML/DL RNN, GAN, Diffusion, NeRF, DIP, Segmentation, High/Low Level SR, Quantization, Parameter Handling

Hobbys

Billiard, Snowboarding, Poetry, Collecting LPs